Energy Market Optimisation Lead (Senior IC)

KAD Agency · Dubai · Fully remote (CET overlap required)

About the role

This is the intelligence layer of the platform — the optimisation system that decides how power moves across bus depots and how distributed energy assets participate in electricity markets. It runs as two levels. The lower level turns a power plan into per-charger decisions in real time. The upper level — this role — builds the plan: it forecasts the market, decides how batteries and charging fleets should bid and operate, and hands the resulting power envelope down to execution.

We are hiring the person who owns the upper level. This is a senior individual-contributor role, not a management one — a hands-on applied-optimisation and operations-research lead who sets the technical direction and ships the code that runs it. The models exist today as research and prototypes; your job is to turn them into production systems that make real decisions about real assets: the modelling approach, the engineering standards, and the path from notebook to service. If you want a role where the hard problems are still open and your decisions stick, this is it.

What you will own

- **The strategic optimiser.** The rolling multi-horizon economic-dispatch model that plans BESS and EV charging across the day against prices, tariffs, and physical limits. Mathematical optimisation — LP, MILP, convex, moving toward stochastic and scenario-based — is the core, and it is yours.
- **The full value stack.** Co-optimise every revenue and saving a battery and depot can reach, not one market in isolation: behind-the-meter services — peak shaving, spot-price and energy arbitrage, and PV self-consumption — together with wholesale day-ahead and intraday participation and the ancillary/balancing-services markets (FCR-N/-D, aFRR, mFRR). The value is in stacking them correctly under one objective, and in aggregating assets toward a virtual power plant (VPP).
- **Market strategy.** Translate each market's rules into constraints and produce the bid recommendations the trading layer acts on. You own the revenue logic and the shadow prices that value flexibility across the stack.
- **Forecasting.** Price, load, and flexibility-availability models with honest, probabilistic confidence and scenario generation. A wrong forecast has a price, so calibration, backtesting, and validation are part of the job, not an afterthought.
- **Co-optimisation.** BESS and depot charging planned in one system rather than two — the competitive edge of the platform. You own proving and holding that advantage.
- **The handoff to execution.** The power envelope handed to the lower level, and the bid recommendations handed to the trading layer, on clean versioned contracts at a fixed cadence.
- **Operational health of your models.** MLOps done properly — backtesting before anything touches production, monitoring and drift detection after, and the judgement call to pull a model from the loop when it misbehaves.

EU standards and compliant formats — mandatory

This is an AI system that influences bids and operations of real energy assets connected to real markets, so it is held to the applicable EU standards, not a best-effort posture:

- **EU AI Act** — governed, explainable optimisation. Every decision must be attributable to its inputs and parameters; no black-box models in production.
- **Reproducibility and traceability** — every model version is accounted for and every decision can be traced and audited. Models are code and are treated that way.
- **GDPR** (EU 2016/679) — asset, site, and driver data classified and handled by design; no PII in logs.
- **NIS2** (Swedish transposition) — the system sits in scope as energy/transport-critical.
- **Data residency: EU only** (europe-north2). No data egress; any external market-data source needs a signed DPA before it goes live.
- **Nordic grid and market rules** — the optimiser must encode the Nordic market structure correctly: day-ahead and intraday (Nord Pool, Elbas), and ancillary services (FCR-N/-D pay-as-marginal, aFRR, mFRR) under SvK requirements. Grid safety always wins: the edge safety layer holds an absolute veto over any optimisation output.

Conformance to these is mandatory, not optional. An optimisation or bid recommendation that is not standards-conformant, reproducible, and explainable is a defect, not a feature gap.

Full stochastic (scenario-based) optimisation is on the horizon — experience building toward it carries real weight here.

Tech stack

Area	Technologies
Language	**Python** (primary) + scientific stack (NumPy, pandas)
Optimisation	MILP / convex / stochastic via Pyomo + HiGHS (CVXPY, Gurobi background welcome)

Area	Technologies
Forecasting / ML	LightGBM, XGBoost, modern time-series stack; Optuna (HPO), Evidently AI (drift)
MLOps	Vertex AI (pipelines, model registry, serving); backtesting and shadow-mode discipline
Platform	GKE (europe-north2), gRPC (internal), FastAPI
Data	BigQuery (feature store, historical market data), TimescaleDB, GCP Pub/Sub; Nord Pool / ENTSO-E feeds
CI/CD & secrets	GitHub Actions, Workload Identity Federation, Secret Manager + External Secrets Operator

What we are looking for

Must-have

- 8+ years in operations research, applied optimisation, or ML engineering, including production ownership of systems you designed.
- Depth in mathematical optimisation — LP, MILP, convex, stochastic — applied to real operational or market problems, with Pyomo, HiGHS, CVXPY, Gurobi, or similar. This is the core of the role.
- Production time-series forecasting: electricity prices, load, or comparable high-stakes signals, including probabilistic and scenario-based methods and gradient-boosted models (LightGBM, XGBoost).
- Strong production Python and the scientific stack, plus solid software-engineering practice — testing, code review, CI. Models are code.
- Experience taking ML from notebook to production on a major cloud, GCP preferred, with real MLOps ownership.
- Enough Nordic energy-market literacy to model the whole value stack correctly: behind-the-meter services (peak shaving, spot/energy arbitrage, PV self-consumption), wholesale day-ahead and intraday (Nord Pool), and ancillary/balancing services (FCR-N/-D, aFRR, mFRR) with pay-as-marginal clearing.
- Backtesting and validation discipline — you prove a strategy against historical market data before it touches production.
- Strong written English; design decisions here are made and defended in writing.

Strong advantage

- Optimising BESS or EV charging operations at asset scale, or building toward a VPP / aggregated flexibility.
- Stochastic or scenario-based optimisation; bi-level or hierarchical optimisation and co-optimisation.
- Balancing-service-provider or market-participation experience (Nord Pool day-ahead/intraday, ENTSO-E data, SvK).
- Vertex AI (pipelines, model registry, serving), BigQuery, TimescaleDB in anger.
- Comfort in a regulated, high-stakes domain where reproducibility and traceability are non-negotiable.

Non-negotiables

- **Grid safety is absolute.** The edge safety layer vetoes any output that would breach a physical limit, and you never design around it. Safety is not a tunable objective.
- **Market commitments are honoured.** What was bid to the grid gets delivered. The plan owns the consequences of its own bids.
- **Backtest before production.** No strategy touches real assets or real bids until it is validated against historical market data.
- **Explainable and auditable by design.** Every model version is accounted for and every decision is traceable to its inputs and parameters. Nothing goes to production that cannot be explained.
- **Contracts are law.** You hand a power envelope to the lower level and bid recommendations to the trading layer on versioned interfaces. You version them properly and never break a consumer without a migration path.